**Summary: Sensitivity Studies**

This Technical Design Report presents a sensitivity analysis of turbojet and turbofan engine performance by varying key component efficiencies. The objective is to identify which components have the most significant impact on Specific Fuel Consumption (SFC) and Specific Thrust ($F_s$), two critical metrics for propulsion system performance.

Using MATLAB-based simulations, the efficiencies of individual components—including the intake, compressor, turbine, burner, nozzle, and fan—were independently varied from -25 % to +25 % of their baseline values (refer to T-20250403-R1-MT-TDR-CUBGT). For both engine types, all other operating conditions such as flight speed, pressure ratio, turbine inlet temperature (TIT), and mechanical efficiency were held constant to isolate the effect of each parameter.

The results are presented as percentage changes in SFC and $F_s$ relative to their baseline values. Comparative plots highlight the sensitivity trends across components. The analysis reveals that certain components, particularly the burner, compressor and turbine efficiencies—exert a much stronger influence on both SFC and $F_s$ than others, such as the intake or fan nozzle efficiency. Differences in sensitivity behaviour between the turbojet and turbofan configurations are also discussed, providing insight into design priorities for different propulsion architectures.

This report serves as a decision-making aid for prioritizing design improvements and optimizing overall engine efficiency, especially in conceptual and preliminary design stages.

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# Nomenclature

| **Symbol** | **Description** | **Units** |
|---|---|---|
| TIT | Turbine Inlet Temperature | K |
| $P_{o_x}$ | Stagnation Pressure at state point x | kPa |
| $T_{o_x}$ | Stagnation Temperature at state point x | K |
| $P_x$ | Static Pressure at state point x | kPa |
| $T_x$ | Static Temperature at State Point x | K |
| $V_x$ | Velocity at state point x | m/s |
| $A_x$ | Area at state point x | $m^2$ |
| $\rho_x$ | Density at state point x | $kg/m^3$ |

| | | |
|---|---|---|
| $h_x$ | Specific Enthalpy at state point x | J/kg |
| $Q$ | Rate of Heat Transfer | J/s |
| $W$ | Rate of Work Transfer | W |
| $\dot{m}$ | Mass Flow Rate | kg/s |
| $\eta$x | Component Isentropic Efficiency | -- |
| LHV | Lower Heating Value | J/kg |
| CPR | Compressor Pressure Ratio | -- |

# 1.0 Introduction

In gas turbine engine design, optimizing performance requires a deep understanding of how individual component efficiencies affect overall system behaviour. Among the most critical metrics used to evaluate engine performance are **(SFC)** and $\boldsymbol{F_s}$ both of which are directly influenced by component-level characteristics such as compressor efficiency, turbine efficiency, and burner performance. Understanding the relative impact of each component is essential for guiding design improvements, setting manufacturing tolerances, and identifying trade-offs in early-stage engine architecture decisions.

This report presents a sensitivity analysis of two engine configurations—a **turbojet** and a **turbofan**—with the goal of determining which component efficiencies most strongly influence SFC and $F_s$. The engine models used for this analysis are based on reverse-engineered baseline performance values obtained from previous technical work. These include values for pressure ratios, turbine inlet temperature (TIT), bypass ratio (for the turbofan), and component efficiencies, all of which reflect realistic estimates of small-scale turbojet and turbofan systems.

The sensitivity study systematically varies the efficiencies of individual components across a range of ±25 % from their baseline values, while holding all other parameters constant. This isolates the effect of each efficiency term on the engine's thrust and fuel consumption characteristics. The results are visualized through MATLAB-generated plots, which reveal the relative importance of each parameter and help identify the components that most significantly affect performance.

By comparing trends across both engine types, this analysis provides valuable insights into which subsystems warrant the greatest attention during design and optimization. The findings serve as a technical foundation for future iterations of the engine design, particularly in enhancing fuel efficiency and thrust output within physical and operational constraints.

# 2.0 Methodology

The sensitivity analysis was performed on two engine configurations—a turbojet and a turbofan—using MATLAB simulations. Baseline performance parameters, such as pressure ratios, component efficiencies, TIT, and bypass ratio (for the turbofan), were derived from prior reverse engineering work.

For each engine, individual component efficiencies were varied independently over a range of -25 % to +25 % relative to their baseline values. The parameters tested include intake, compressor, turbine, burner, nozzle, mechanical, and fan efficiencies (for the turbofan). All other design and environmental conditions were held constant to isolate the effects of each component.

The MATLAB scripts compute changes in SFC and $F_s$ for each variation. These outputs are normalized against their respective baseline values and plotted to visualize sensitivity trends.

## 2.1 Turbojet

The input parameters used for the turbojet engine model are summarized in **Table 1**. These values were obtained through reverse engineering of Jet-Cat P500 turbojet engine and serve as the reference condition for all sensitivity calculations. The parameters include ambient conditions, fuel properties, mass flow rates, and baseline component efficiencies. These baseline values ensure that the sensitivity results are grounded in a realistic and consistent performance model.

| Parameter | Values |
|---|---|
| $\eta_i$ | 0.903 |
| $\eta_c$ | 0.8315 |
| $\eta_b$ | 0.9125 |
| $\eta_t$ | 0.7851 |
| $\eta_j$ | 0.95 |
| $\eta_m$ | 0.99 |
| $f_b$ | 0.037 |

| Fixed | Values |
|---|---|
| $CPR$ | 3.6 |
| $TIT\ (K)$ | 1148.65 |
| $V_0$ | 17.67 m/s |
| $F$ | 500 N |
| $Altitude$ | 0 |
| Mach | 0 |

Table 1: Input parameters for turbojet

### 2.2 Turbofan

The turbofan engine model uses input parameters listed in **Table 2**, same as P500 turbojet. These include ambient conditions, flight speed, pressure ratios, turbine inlet temperature, and baseline component efficiencies. The turbofan model also assumes

a fixed mechanical efficiency, Bypass ratio, FPR, and accounts for both core and bypass flow paths, making it suitable for assessing sensitivity to variations in fan and core nozzle efficiencies as well.

| Parameters | Values |
|---|---|
| $\eta_i$ | 0.903 |
| $\eta_{fan}$ | 0.85 |
| $\eta_c$ | 0.8315 |
| $\eta_b$ | 0.9125 |
| $\eta_{LPT}$ | 0.8 |
| $\eta_{HPT}$ | 0.7851 |
| $\eta_{n_c}$ | 0.9 |
| $\eta_{n_h}$ | 0.95 |

| Fixed | Values |
|---|---|
| $\eta_m$ | 0.99 |
| $TIT$ (K) | 1148.65 |
| $FPR$ | 1.2 |
| CPR | 3.6 |
| OPR | 4.32 |
| BPR | 2 |

Table 2: Input parameters for turbofan

# 3.0 Results & Discussion

The results of the sensitivity analysis are presented through a series of MATLAB-generated plots for both turbojet and turbofan engines. These plots illustrate how variations in individual component efficiencies influence $SFC\ and\ F_s$ Additional parametric analysis was also conducted with respect to Compressor Pressure Ratio (CPR) to identify optimal design points.

### 3.1 Turbojet Engine

Figure 1 shows the variation in SFC as component efficiencies are varied from -25 % to +25 %. Among all parameters, the **burner efficiency** ($\eta_b$) exhibits the **greatest influence on SFC**, followed by the **turbine** and **compressor efficiencies**. As turbine and compressor efficiencies increase, SFC initially decreases; however, beyond a certain point, the SFC begins to rise again due to **choking in the flow**, which limits further gains in performance.

In addition to component efficiency variation, Figure 2 present SFC and $F_s$ trends as functions of **CPR** across various TITs. The results reveal that:

- **Optimal SFC occurs around a CPR of 15**, balancing compression work with fuel usage.
- **Maximum specific thrust occurs near a CPR of 3.5**, aligning closely with the JetCat P500's actual configuration, validating the reverse-engineered model.

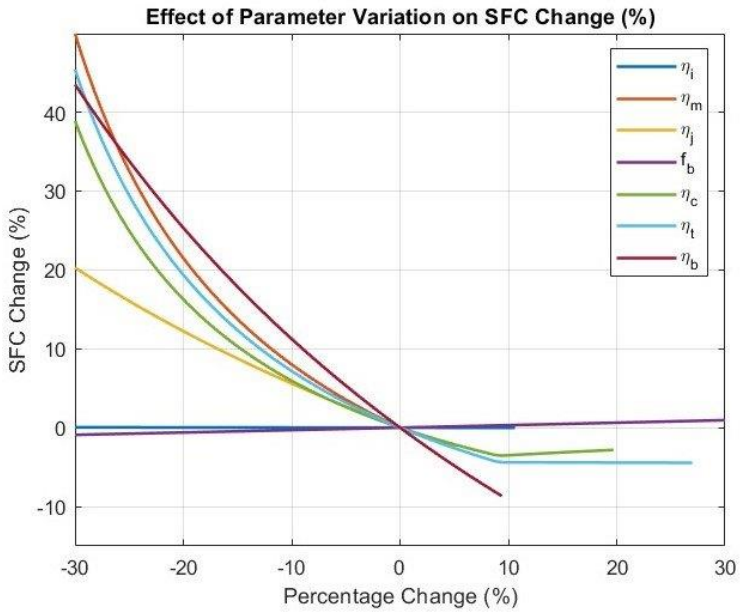


Fig 1: Effect on SFC due to component efficiencies for turbojet

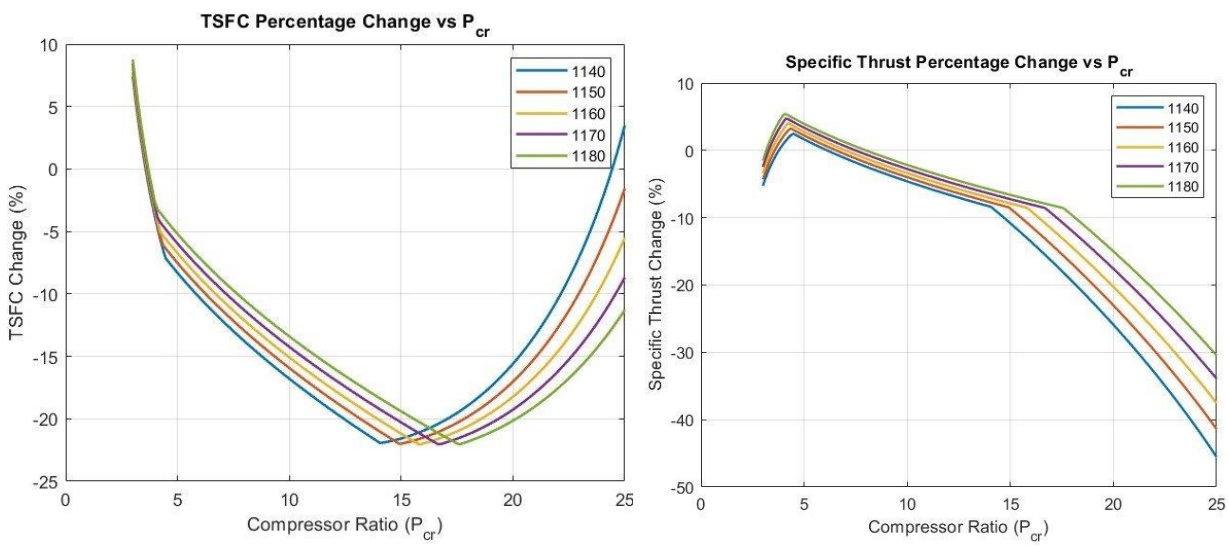


Fig 2: Pressure ratio effect on the SFC and Specific thrust for turbojet

### 3.2 Turbofan Engine

The turbofan sensitivity results, shown in Figures 3 and 4, display trends similar to those observed for the turbojet. Component efficiencies such as **compressor, turbine, burner, and nozzle efficiencies** have the strongest impact on both

SFC and specific thrust. However, due to the fixed **Bypass Ratio (BPR) of 2**, **Fan Pressure Ratio (FPR) of 1.2**, and **Overall Pressure Ratio (OPR) of 3.6**, the baseline values already reflect a lower thrust configuration.

At the reference point (0 % change), the **SFC is approximately 25 % lower** than that of the turbojet, owing to the improved propulsive efficiency of the turbofan. On the other hand, $F_s$ **is significantly reduced—by nearly 60 %**, highlighting the classic trade-off in turbofan design between fuel economy and thrust output. This reduction is expected in low BPRconfigurations where propulsive efficiency improves but momentum thrust decreases.

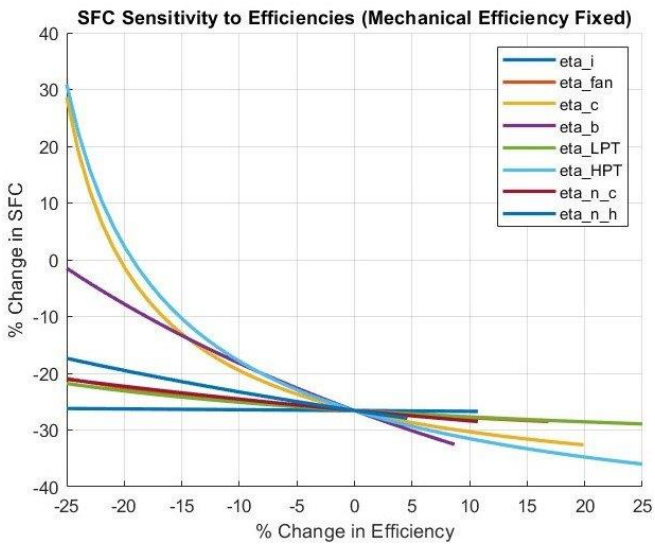


Fig 3: Effect on SFC due to component efficiencies for turbofan

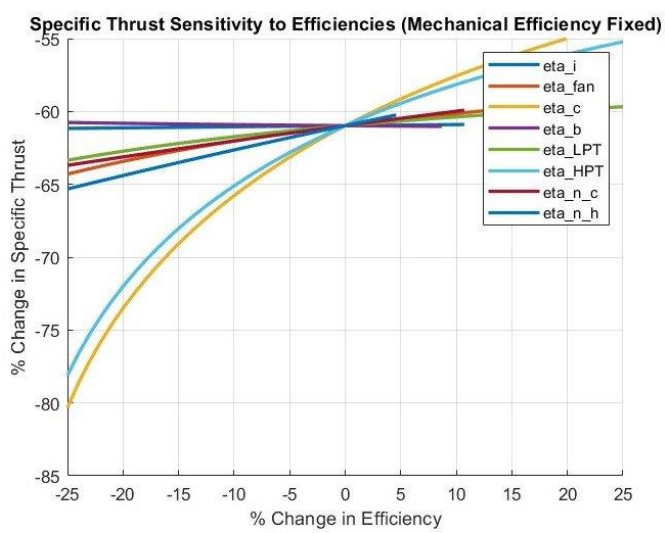


Fig 4: Effect on Specific thrust due to component efficiencies for turbofan

# 4.0 References

[1] H. I. H. Saravanamuttoo, G. F. C. Rogers, H. Cohen, P. Straznicky, and A. Nix, *Gas turbine theory*, 7th ed. Harlow; New York; Toronto Pearson, 2017.

[2] M. J. Moran, H. N. Shapiro, D. D. Boettner, and M. B. Bailey, *Fundamentals of Engineering Thermodynamics, 9th Edition*. Wiley, 2018.

[3] Jet-Cat Germany, "P500-PRO-GL – Jet-Cat," *Jetcat.de*, 2024. https://www.jetcat.de/en/productdetails/produkte/jetcat/produkte/Professionell/P500%20PRO%20-%20GL (accessed Oct. 01, 2024).

# 5.0 Appendix

Turbojet:

```
clc; clear; close all;

% Base values
P_cr = 3.6;
altitude = 0;
M = 0.0;
[T_a, V_a, P_amb] = atmosisa(altitude);
P_a = P_amb / 1000;
mass_air = 0.9;
V_inlet = 17.6679;
LHV = 43260000;
F = 500;

% Core values for efficiencies and TIT
eta_intake = 0.903;
eta_mech = 0.99;
eta_nozzle = 0.95;
pressure_drop = 0.037;
eta_compressor = 0.8315;
eta_turbine = 0.7851;
eta_burner = 0.9125;
TIT = 1148.65;

% Constants
gamma_air = 1.4;
gamma_gas = 1.333;
```

```
Cp_air = 1005;
Cp_gas = 1148;
R = Cp_gas * (gamma_gas - 1) / gamma_gas;

% Define variation range (-25% to +25%)
percent_change = linspace(-30, 30, 100);
scaling_factor = 1 + percent_change / 100;

% Storage for performance metrics
SFC = nan(length(scaling_factor), 8);
SpecificThrust = nan(length(scaling_factor), 8);
ThermalEff = nan(length(scaling_factor), 8);
PropEff = nan(length(scaling_factor), 8);
OverallEff = nan(length(scaling_factor), 8);

% Reference values for percentage change
SFC_ref = 0.151;
SpecificThrust_ref = 546.68;

% Parameters list for legend
params = {'\eta_i', '\eta_m', '\eta_j', 'f_b', ...
          '\eta_c', '\eta_t', '\eta_b'};

for i = 1:7
    for j = 1:length(scaling_factor)
        % Modify parameters with constraints
        eta_intake_var = eta_intake * (scaling_factor(j)^(i == 1));
        eta_mech_var = eta_mech * (scaling_factor(j)^(i == 2));
        eta_nozzle_var = eta_nozzle * (scaling_factor(j)^(i == 3));
        pressure_drop_var = pressure_drop * (scaling_factor(j)^(i == 4));
        eta_compressor_var = eta_compressor * (scaling_factor(j)^(i == 5));
        eta_turbine_var = eta_turbine * (scaling_factor(j)^(i == 6));
        eta_burner_var = eta_burner * (scaling_factor(j)^(i == 7));


        % Skip values beyond the valid range
        if eta_intake_var > 1 || eta_mech_var > 1 || eta_nozzle_var > 1 || ...
           eta_compressor_var > 1 || eta_turbine_var > 1 || eta_burner_var > 1 || ...
           eta_burner_var < 0 || pressure_drop_var < 0
            continue;
        end

        % ENGINE CALCULATIONS

        % Stagnation conditions
        To_1 = T_a + (V_inlet^2) / (2 * Cp_air);
        Po_1 = P_a * (1 + (V_inlet^2) / (2 * Cp_air * T_a))^(gamma_air / (gamma_air - 1));

        % Intake
        To_2 = T_a + (V_inlet^2) / (2 * Cp_air);
```

```
        Po_2 = P_a * (1 + (eta_intake_var * V_inlet^2) / (2 * Cp_air * T_a))^(gamma_air /
(gamma_air - 1));

        % Compressor
        To_3 = To_2 + To_2 / eta_compressor_var * ((P_cr)^(1 / (gamma_air / (gamma_air -
1))) - 1);
        Po_3 = P_cr * Po_2;

        % Combustor
        Po_4 = Po_3 * (1 - pressure_drop_var);
        To_4 = TIT;
        energy_input = Cp_air * (To_3 - To_2);

        % Turbine
        To_5 = To_4 - (energy_input) / (Cp_gas * eta_mech_var);
        To_5s = To_4 - (To_4 - To_5) / eta_turbine_var;
        Po_5 = Po_4 * (To_5s / To_4)^(gamma_gas / (gamma_gas - 1));

        % Nozzle
        nozzle_pressure_ratio = Po_5 / P_a;
        critical_pressure_ratio = 1 / (1 - (1 / eta_nozzle_var) * ((gamma_gas - 1) /
(gamma_gas + 1)))^(gamma_gas / (gamma_gas - 1));

        if nozzle_pressure_ratio > critical_pressure_ratio
            T_6 = (2 / (gamma_gas + 1)) * To_5;
            P_6 = Po_5 / critical_pressure_ratio;
            rho_6 = (P_6 * 100) / (R / 1000 * T_6);
            velocity_exit = sqrt(gamma_gas * R * T_6);
            A_6_by_m = (rho_6 * velocity_exit)^(-1);
        else
            P_6 = P_a;
            T_6 = To_5 - (eta_nozzle_var * To_5 * (1 - (1 /
nozzle_pressure_ratio)^((gamma_gas-1) / gamma_gas)));
            velocity_exit = sqrt(2 * Cp_gas * (To_5 - T_6));
            rho_6 = (P_6 * 100) / (R / 1000 * T_6);
            A_6_by_m = (rho_6 * velocity_exit)^(-1);
        end

        % Fuel-Air Ratio
        fuel_air_ratio = (Cp_gas * TIT - Cp_air * To_3) / (eta_burner_var * LHV - Cp_gas *
TIT);

        % Specific Thrust
        specific_thrust = (velocity_exit * (1 + fuel_air_ratio) - V_inlet) + ((P_6 - P_a) *
(A_6_by_m) * 10^3);

        % TSFC Calculation
        m_air = F / specific_thrust;
        m_fuel = m_air * fuel_air_ratio;
        TSFC = (m_fuel * 3600) / F;
```

```
        % Efficiency Calculations
        thermal_efficiency = (m_air * (velocity_exit^2 - V_inlet^2) / 2) / (m_fuel * LHV);
        propulsive_efficiency = 2 * V_inlet * ((1 + fuel_air_ratio) * velocity_exit -
V_inlet) / ((1 + fuel_air_ratio) * velocity_exit^2 - V_inlet^2);
        overall_efficiency = thermal_efficiency * propulsive_efficiency;

        % Store results
        SFC(j, i) = (TSFC - SFC_ref) / SFC_ref * 100;
        SpecificThrust(j, i) = (specific_thrust - SpecificThrust_ref) / SpecificThrust_ref
* 100;
        ThermalEff(j, i) = thermal_efficiency * 100;
        PropEff(j, i) = propulsive_efficiency * 100;
        OverallEff(j, i) = overall_efficiency * 100;
    end
end

% Plot results
plot_labels = {'SFC Change (%)', 'Specific Thrust Change (%)', 'Overall Efficiency (%)',
...
               'Propulsive Efficiency (%)', 'Thermal Efficiency (%)'};
data_sets = {SFC, SpecificThrust, OverallEff, PropEff, ThermalEff};

for k = 1:length(data_sets)
    figure;
    hold on; grid on; box on;
    for i = 1:7
        valid_idx = ~isnan(data_sets{k}(:, i)); % Remove invalid points
        plot(percent_change(valid_idx), data_sets{k}(valid_idx, i), 'LineWidth', 1.5,
'DisplayName', params{i});
    end
    xlabel('Percentage Change (%)');
    ylabel(plot_labels{k});
    title(['Effect of Parameter Variation on ', plot_labels{k}]);
    legend('show', 'Location', 'best');

    % Set y-axis limit for SFC plot
    if k == 1
        ylim([-15 inf]); % Limit SFC change plot to start from -25%
    end
end
```

Turbofan:

```
clc;
clear;

% TURBINE INLET TEMPERATURE:
To_4 = 1200; % HPT Inlet Temperature
```

```
% Input Parameters
[T, a, P, rho] = atmosisa(1000); % ISA conditions at 1000 m
P_a = P / 1000; % Ambient Pressure [kPa]
T_a = T; % Ambient Temperature [K]
V_inf = 75; % Flight Speed [m/s]

% Fixed Parameters
F_total = 500; % Thrust [N]
BPR = 2; % Bypass Ratio
FPR = 1.2; % Fan Pressure Ratio
OPR = 4.32; % Overall Pressure Ratio

% Constants
C_p_air = 1005; % J/kg*K
Gamma_air = 1.4;
C_p_gas = 1148; % J/kg*K
Gamma_gas = 1.333;
R = (C_p_air * (Gamma_air - 1) / Gamma_air) / 1000; % kJ/kg*K
LHV = 43260000; % J/kg
d_pb = 0.037; % Combustor Pressure Loss

% Core Efficiency Parameters (baseline)
core_efficiencies = struct( ...
    'eta_i', 0.903, ...
    'eta_fan', 0.85, ...
    'eta_c', 0.8315, ...
    'eta_b', 0.9125, ...
    'eta_LPT', 0.8, ...
    'eta_HPT', 0.7851, ...
    'eta_n_c', 0.9, ...
    'eta_n_h', 0.95);

% Fixed mechanical efficiency
eta_m_fixed = 0.99;

% Core values for comparison
SFC_core = 0.151; % kg/N.hr
specific_thrust_core = 546.46; % m/s
rho = 1.225; %kg/m^3

% Efficiency fields to vary
eff_names = fieldnames(core_efficiencies);

% Variation range (-25% to +25%)
variation = linspace(-0.25, 0.25, 50);

% Store results
results = struct();
```

```
for e = 1:length(eff_names)
    eff_name = eff_names{e};
    base_value = core_efficiencies.(eff_name);
    SFC_arr = [];
    ST_arr = [];
    eff_var = [];

    for i = 1:length(variation)
        eff_factor = 1 + variation(i);
        new_eff = base_value * eff_factor;

        % Skip if efficiency > 1 or <= 0
        if new_eff > 1 || new_eff <= 0
            continue;
        end

        % Assign current efficiencies
        eta_i = core_efficiencies.eta_i;
        eta_fan = core_efficiencies.eta_fan;
        eta_c = core_efficiencies.eta_c;
        eta_b = core_efficiencies.eta_b;
        eta_LPT = core_efficiencies.eta_LPT;
        eta_HPT = core_efficiencies.eta_HPT;
        eta_n_c = core_efficiencies.eta_n_c;
        eta_n_h = core_efficiencies.eta_n_h;
        eta_m = eta_m_fixed; % Fixed mechanical efficiency

        % Update current parameter
        eval([eff_name ' = new_eff;']);

        %% ---- Intake ----
        P_1 = P_a;
        T_1 = T_a;
        To_2 = T_1 + (V_inf^2) / (2 * C_p_air);
        Po_2 = P_1 * ((1 + eta_i * (V_inf^2 / (2 * C_p_air * T_a)))^(Gamma_air / (Gamma_air
- 1)));

        %% ---- Fan ----
        gamma_air = 1.4;
        gamma_air_exp = (gamma_air - 1) / gamma_air;
        Po_3_prime = FPR * Po_2;
        To_3_prime = To_2 * (1 + (FPR^gamma_air_exp - 1) / eta_fan);

        %% ---- Compressor ----
        Po_3 = (OPR / FPR) * Po_3_prime;
        To_3 = To_3_prime * (1 + ((OPR / FPR)^gamma_air_exp - 1) / eta_c);

        %% ---- Combustor ----
        Po_4 = Po_3 * (1 - d_pb);
```

```
        %% ---- High Pressure Turbine ----
        gamma_gas = 1.333;
        gamma_gas_exp = gamma_gas / (gamma_gas - 1);
        To_5_prime = To_4 - ((C_p_air * (To_3 - To_3_prime)) / (C_p_gas * eta_m));
        To_5s_prime = To_4 - ((To_4 - To_5_prime) / eta_HPT);
        Po_5_prime = Po_4 * (To_5s_prime / To_4)^(gamma_gas_exp);

        %% ---- Low Pressure Turbine ----
        To_5 = To_5_prime - (BPR + 1) * (C_p_air / (eta_m * C_p_gas)) * (To_3_prime -
To_2);
        To_5s = To_5_prime - ((To_5_prime - To_5) / eta_LPT);
        Po_5 = Po_5_prime * (To_5s / To_5_prime)^(gamma_gas_exp);

        %% ---- Fuel to Air Ratio ----
        f = ((C_p_gas * To_4) - (C_p_air * To_3)) / ((eta_b * LHV) - (C_p_gas * To_4));

        %% ---- Nozzle Cold Flow (Fan Stream) ----
        CPR_Cold = (1) / ((1 - ((1 / eta_n_c) * ((Gamma_air - 1) / (Gamma_air +
1))))^(Gamma_air / (Gamma_air - 1)));
        T6_prime = To_3_prime - (eta_n_c * To_3_prime * (1 - ((1 / FPR)^((Gamma_air - 1) /
Gamma_air))));
        V6_prime = sqrt(2 * C_p_air * (To_3_prime - T6_prime));

        %% ---- Core Nozzle ----
        NPR_Hot = Po_5 / P_a;
        CPR_Hot = (1) / ((1 - ((1 / eta_n_h) * ((Gamma_gas - 1) / (Gamma_gas +
1))))^(Gamma_gas / (Gamma_gas - 1)));
        T6 = To_5 - (eta_n_h * To_5 * (1 - ((1 / NPR_Hot)^((Gamma_gas - 1) / Gamma_gas))));
        V6 = sqrt(2 * C_p_gas * (To_5 - T6));

        %% ---- Mass Flow ----
        m_total = (F_total * (1 + BPR)) / (V6 * (1 + f) + V6_prime * BPR - V_inf * (1 +
BPR));
        m_hot = m_total / (BPR + 1);
        m_fuel = m_hot * f;

        %% ---- Specific Thrust & SFC ----
        specific_thrust = F_total / m_total;
        SFC = m_fuel * 3600 / F_total;
        Area = m_total/(rho*V_inf);

        % Store
        eff_var(end+1) = (eff_factor - 1) * 100; % % change
        ST_arr(end+1) = (specific_thrust - specific_thrust_core) / specific_thrust_core *
100;
        SFC_arr(end+1) = (SFC - SFC_core) / SFC_core * 100;
    end

    % Store results
    results.(eff_name).eff_change = eff_var;
```

```
    results.(eff_name).SFC_change = SFC_arr;
    results.(eff_name).ST_change = ST_arr;
end

%% ---- Single Plot for SFC ----
figure;
hold on;
fields = fieldnames(results);
for f = 1:length(fields)
    plot(results.(fields{f}).eff_change, results.(fields{f}).SFC_change, 'LineWidth', 2,
'DisplayName', strrep(fields{f}, '_', '\_'))
end
xlabel('% Change in Efficiency');
ylabel('% Change in SFC');
title('SFC Sensitivity to Efficiencies (Mechanical Efficiency Fixed)');
legend show;
grid on;
hold off;

%% ---- Single Plot for Specific Thrust ----
figure;
hold on;
for f = 1:length(fields)
    plot(results.(fields{f}).eff_change, results.(fields{f}).ST_change, 'LineWidth', 2,
'DisplayName', strrep(fields{f}, '_', '\_'))
end
xlabel('% Change in Efficiency');
ylabel('% Change in Specific Thrust');
title('Specific Thrust Sensitivity to Efficiencies (Mechanical Efficiency Fixed)');
legend show;
grid on;
hold off;
```